IITISoC 2026

Attitude Control of a Satellite Using Reaction Wheels

Mid Evaluation Report

Team Members: [B.NANDINI]
[M.SREE CHARAN]

Department: Mechanical Engineering
Institute: Indian Institute of Technology Indore
Program: B.Tech, Second Year

GitHub: https://github.com/me250003020-dot/Satellite-Attitude-Control-Using
Google Drive: https://drive.google.com/drive/folders/1s3tD5Q_gKMdLsJk-uZaypLn5x9_

July 31, 2026

Contents

1 INTRODUCTION

This Report is a documentation of our complete solution to PS 2, which is reaction wheel-based attitude control of satellites. We chose Fusion 360 to design a 3U CubeSat model and Simulink and MATLAB for simulating the system. The system has been validated under both of the required scenarios - scenario A (without any external disturbance) and scenario B (with disturbances). We have taken the angles and values considering the satellite's frame as the frame of reference.

2 PROBLEM STATEMENT ANALYSIS

2.1 Objective

Core requirements of the PS were to design,model and simulate reaction wheel based control system for a cubesat The extension we made in the final solution was addition of a mechanism to recover from wheel saturation using an additional actuator type

2.2 Workflow

We began by understanding the PS and the physics concepts required to solve it. Our team then started to research about the applications that would help us approach the problem statement efficiently,we compared different softwares available for different parts of the solution. Then we started to work with CAD 3U satellite designing and with those derived inertia values, we derived equations of motion. Then we proceeded into the controls part of the problem statement; we chose a suitable controller, designed the circuit , then tuned the controllers and made a simulation of single-axis attitude control by Mid-Evaluation. After Mid-Evaluation, the three axises where individually tested to check the consistency of our system, and introduced a few more different kind disturbances were added; then, an actuator subsytem was introduced replacing the intial saturation block, Furthermore,the equations were modified again, considering the coupling terms which werent there earlier in single-axis simulation. This shift of equations integrated the system of 3 axes. The PID and PI controllers were tuned to acquire the correct automatic attitude controlling system in both Scenario A and Scenario B

3 CAD DESIGN

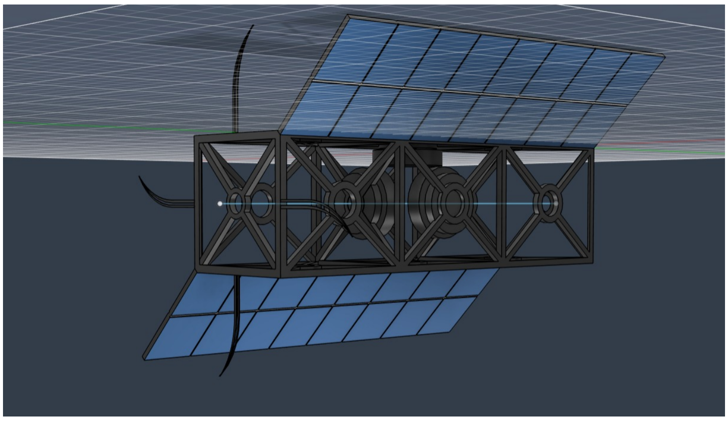

Figure 1: CubeSat assembly with three mutually perpendicular reaction wheels.

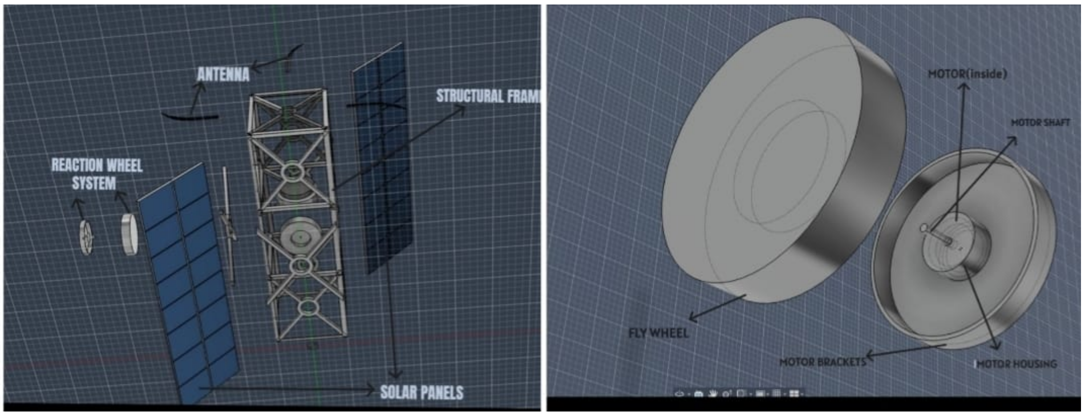

Figure 2: Exploded view of CubeSat assembly

REASONING FOR CHOICES IN DESIGN:

The structural frame was designed with a central circular ring connected to the corner members through truss elements to achieve a balance between rigidity, symmetry, and weight reduction. This circular ring acts as a structural hub, providing a continuous load path and improving stiffness while avoiding the stress concentrations associated with sharp corners.

This design also provides a convenient and robust framework for integrating internal subsystems while remaining consistent with lightweight

Solar panels and antennas were included to closely replicate the external configuration of a practical CubeSat, making the CAD model more representative of an actual spacecraft.

The motor housing in our design was made circular in shape because of 3 reasons:
1.This Provides uniform support around the cylindrical motor housing.
2.It Improves alignment between the motor shaft and reaction wheel.
3.Distributes the clamping load more evenly, reducing local stress concentrations.
4.Helps in minimising small mechanical misalignments and unwanted vibrations during wheel rotation.

DERIVED PHYSICAL PARAMETERS:

Mass and inertia values were extracted directly from the completed Fusion 360 assembly using the Physical Properties tool, evaluated relative to each body's center of mass, with steel/aluminum material properties assigned.

Table 1: Satellite frame inertia parameters (from CAD)

Parameter	Value
Mass	1.40621 kg
I_{xx}	0.01819 kg·m^2
I_{yy}	0.01238 kg·m^2
I_{zz}	0.01819 kg·m^2

Table 2: Reaction wheel inertia parameters (from CAD, single wheel)

Parameter	Value
Mass	0.06975 kg
I_{xx} (mounting/spin axis)	2.8491×10^{-5} kg·m^2

DESIGN CONSIDERATIONS:

Our priorities were:
1. Mass distribution realism
Placing components such that satellite's center off mass remained closer to the it's geometric center . This simplifies the moment of inertia tensor for the latter purposes in project
2 .Modularity
Subsystems were initially developed as independent components and then combined later, so that making changes in subsystems doesn't require rebuilding the entire model.
3. Manufacturability
The system was developed considering some important industrial manufacturing restraints such as repetitions of components , symmetric placements etc.

4 PHYSICS INVOLVED

4.1 Reaction Wheel Principle

The attitude control system works on the principle of conservation of angular momentum. When a reaction wheel accelerates or decelerates, an equal and opposite torque is generated on the satellite body, causing it to rotate about the corresponding axis. Three independent reaction wheels are used to control roll, pitch and yaw.

$$H = I_s\omega_s + I_w\omega_w \tag{1}$$

$$\frac{dH}{dt} = 0 \tag{2}$$

4.2 Attitude Error

The controller continuously compares the desired attitude with the actual attitude of the satellite to calculate the error for each axis.

$$e = \theta_{\text{desired}} - \theta_{\text{actual}} \tag{3}$$

$$e = \begin{bmatrix} e_{\text{roll}} \\ e_{\text{pitch}} \\ e_{\text{yaw}} \end{bmatrix} \tag{4}$$

4.3 Rigid Body Dynamics

The rotational motion of the satellite is modelled using Euler's rigid body equations including gyroscopic coupling.

$$I\dot{\omega} + \omega \times (I\omega) = T \tag{5}$$

$$\dot{\omega} = I^{-1} \left[T - \omega \times (I\omega) \right] \tag{6}$$

$$\omega = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \tag{7}$$

4.4 DC Motor Electrical Dynamics

The motor electrical behaviour was modelled using the standard armature voltage equation.

$$V = Ri + L\frac{di}{dt} + K_e\omega_w \tag{8}$$

$$\frac{di}{dt} = \frac{V - Ri - K_e\omega_w}{L} \tag{9}$$

This captures the influence of armature resistance, inductance and back electromotive force on motor current.

4.5 Disturbances In Scenario B

Scenario B incorporates major disturbance torques acting on the spacecraft.

Total disturbance,

$$T_d = T_g + T_{\text{drag}} + T_{\text{SRP}} + T_{\text{noise}} + T_{\text{constant}} \tag{10}$$

Gravity Gradient Torque

$$T_g = \frac{3\mu}{r^3}(I_{\max} - I_{\min})\sin(2\theta) \tag{11}$$

Aerodynamic Drag

$$F_d = \frac{1}{2}\rho C_d A V^2 \tag{12}$$

$$T_{\text{drag}} = F_d L \tag{13}$$

Solar Radiation Pressure

$$P = 4.56 \times 10^{-6}\ \text{N/m}^2 \tag{14}$$

$$T_{\text{SRP}} = Fr \tag{15}$$

White noise was included to represent measurement uncertainty.

4.6 Kinematics

Angular velocity is integrated to obtain attitude:

$$\dot{\phi} = p + q \sin\phi \, \tan\theta + r \cos\phi \, \tan\theta \tag{16}$$

$$\dot{\theta} = q \cos\phi - r \sin\phi \tag{17}$$

$$\dot{\psi} = \frac{q \sin\phi + r \cos\phi}{\cos\theta} \tag{18}$$

The angular velocities obtained from the dynamics are integrated to determine the satellite orientation.

These Euler angles are continuously fed back to the controller.

4.7 Inertia Tensor

The mass distribution obtained from the CAD model was represented using the inertia tensor. The inertia matrix governs the rotational response of the satellite about its principal axes and is directly used in Euler's rigid body equations.

$$I = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{bmatrix} \tag{19}$$

For the developed CubeSat, the centre of mass was intentionally maintained close to the geometric centre during CAD modelling to minimise the off-diagonal terms of the inertia matrix.

4.8 Angular Momentum

The spacecraft angular momentum is obtained from the inertia tensor and angular velocity vector.

$$H = I\omega \tag{20}$$

$$\omega = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \tag{21}$$

This relation is used while formulating the rotational equations of motion.

4.9 Torque To Current Conversion

The control torque generated by the PID controller is converted into the required motor current using the motor torque constant.

$$T = K_t i \tag{22}$$

$$i_{\text{ref}} = \frac{T}{K_t} \tag{23}$$

This forms the reference current for the inner current control loop.

4.10 Control Law: PI

A PI controller regulates the motor current so that the actual motor current tracks the reference current generated from the attitude controller.

Current error,

$$e_i = i_{\text{ref}} - i \tag{24}$$

PI controller,

$$V = K_p e_i + K_i \int e_i \, dt \tag{25}$$

This inner loop provides faster electrical response while allowing the outer PID loop to focus on attitude regulation.

4.11 Control Law: PID

A proportional-integral-derivative controller drives the system:

$$\tau_{command}(t) = K_p \, e(t) + K_i \int_0^t e(\tau) \, d\tau + K_d \, \dot{e}(t), \qquad e(t) = \theta_{desired} - \theta_{actual}(t) \tag{26}$$

The proportional term provides immediate correction, the integral term removes steady-state error, and the derivative term reduces overshoot and oscillations.
JUSTIFICATION FOR PID USAGE:
PID was selected because it provides a good balance between accuracy, stability and implementation simplicity. Compared to P, PI and PD controllers, it achieves better pointing performance while remaining easy to tune and implement in Simulink compared to advanced controllers such as LQR or Model Predictive Control.

5 SIMULINK IMPLEMENTATION

5.1 Flowchart

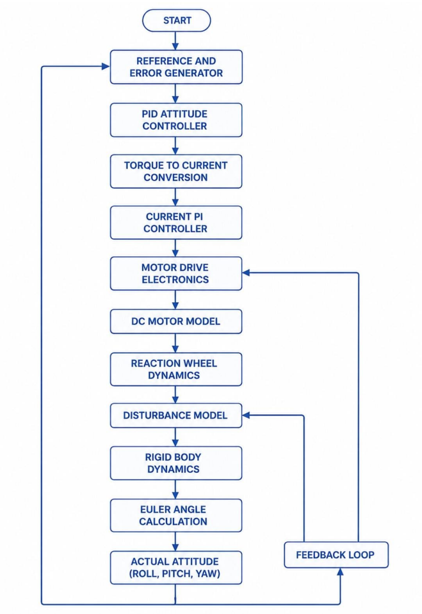

Figure 3: Flow Of Signal In System

5.2 Block-by-Block Description

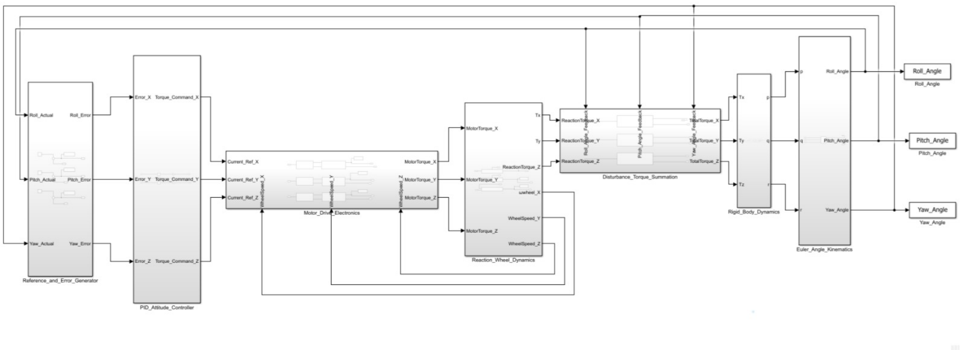

Figure 4: Complete 3-axis closed-loop Simulink architecture, as implemented.

1. REFERENCE AND ERROR GENERATOR:
Purpose-
Generates the desired attitude commands and computes the corresponding roll, pitch and yaw errors using the measured spacecraft attitude.

Implementation-
Reference angles are provided independently for all three axes. These are continuously compared with the feedback angles received from the dynamics model, producing the error signals that drive the attitude controller.

2. PID (ATTITUDE CONTROLLER) SUBSYSTEM:
Purpose-
Generates the control torque required to minimise the attitude error for each rotational axis.

Implementation-
Independent PID controllers are implemented for roll, pitch and yaw. Each controller processes its respective error signal and produces the torque command that is passed to the actuator subsystem.

3. MOTOR DRIVE ELECTRONICS:
Purpose-
This subsystem represents the complete actuator model of the spacecraft. Instead of applying the commanded torque directly, the actuator behaviour is modelled by incorporating the motor drive electronics and DC motor dynamics.

Internal Subsystems:
(Torque to Current Reference) Converts the commanded control torque into the corresponding motor current reference required by the actuator.

Current PI Controller-
Regulates the motor current by continuously adjusting the applied motor voltage. The inner current loop ensures fast actuator response while accurately tracking the commanded current.
DC Motor-
Models the electrical behaviour of the reaction wheel motor and generates the motor torque supplied to the reaction wheel assembly. This makes the actuator behaviour significantly more

realistic than using an ideal torque source.

4. REACTION WHEEL DYNAMICS:
Purpose-
Models the mechanical behaviour of the reaction wheel assembly.
Implementation-
The motor torque generated by the actuator accelerates or decelerates the reaction wheel. The corresponding reaction torque produced on the spacecraft is calculated and forwarded to the spacecraft dynamics, while the wheel speed is simultaneously monitored.

5. DISTURBANCE TORQUE SUMMATION:
Purpose-
Combines all external disturbance torques acting on the spacecraft before they are applied to the dynamics model.
Implementation-
For Scenario B, gravity gradient torque, aerodynamic drag, solar radiation pressure and Gaussian noise are summed with the control torque to simulate realistic orbital disturbances.

6. RIGID BODY DYNAMICS:
Purpose-
Calculates the rotational response of the spacecraft.
Implementation-
The subsystem receives the net applied torque from the controller and disturbance model and computes the resulting angular motion of the satellite. The generated angular velocities are forwarded to the kinematics subsystem for attitude calculation.

7. EULER ANGLE KINEMATICS:
Purpose-
Determines the spacecraft attitude from the calculated angular velocities.
Implementation-
The angular velocity components are integrated to obtain the roll, pitch and yaw angles. These updated attitude values are then fed back to the Reference and Error Generator, completing the closed-loop control system.

6 CONTROLLER CONFIGURATIONS

The PID Controller block is configured with the gains arrived at through the tuning are as follows : X-axis

$$K_p = 0.015, \quad K_i = 0.002, \quad K_d = 0.045 \tag{27}$$

Y-axis

$$K_p = 0.0544, \quad K_i = 0.0061, \quad K_d = 0.05 \tag{28}$$

Z-axis

$$K_p = 0.015, \quad K_i = 0.002, \quad K_d = 0.045 \tag{29}$$

This set produces a stable, well-damped response with a peak wheel speed within a realistic range for a small CubeSat reaction wheel motor, as shown in the results below.

7 RESULTS : SCENARIO A

The satellite was commanded from an initial Roll of 0° to a target of 45° , Pitch of 0° to 30°, Yaw of 0° to 45° with no external disturbance.

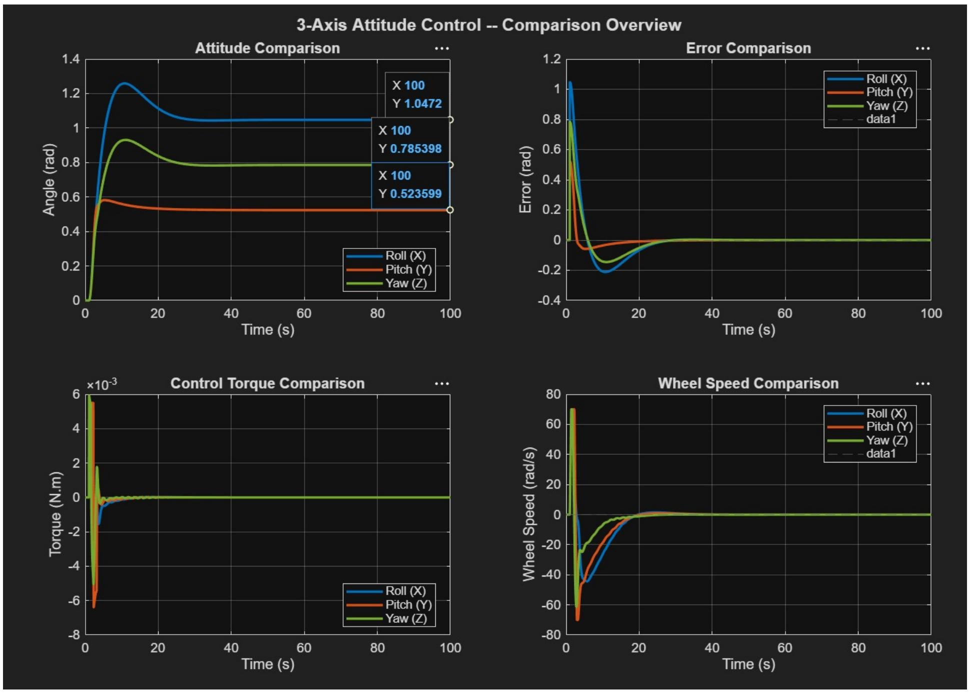

Figure 5: Scenario A: satellite attitude, attitude error, reaction wheel speed, and control torque vs. time.

================= 3-AXIS SIMULATION RESULTS =================

Metric	Roll (X)	Pitch (Y)	Yaw (Z)
Desired Angle (deg)	60.000	30.000	45.000
Rise Time (s)	3.327	1.207	3.466
Peak Time (s)	10.793	5.223	11.113
Overshoot (%)	20.17	11.16	18.54
Settling Time (s)	24.353	18.843	24.703
Steady State Err (rad)	0.000000	-0.000000	0.000000
RMS Error (rad)	0.159400	0.061908	0.114131
Max Torque (N.m)	0.005933	0.005933	0.005933
Max Wheel Speed(rad/s)	70.000000	70.000000	70.000000

Figure 6: Scenario A: Performance Metrics (3-AXES)

Discussion: The system successfully reached the desired roll, pitch and yaw angles with zero steady-state error in all three axes. Rise times varied between 1.21 s and 3.47 s, while the responses settled within 18.84–24.35 s. The controller produced a moderate overshoot of 11–20percent, indicating a good balance between response speed and stability. The maximum control torque remained limited to 0.0059 N·m, and the reaction wheel speed reached the imposed limit of 70 rad/s, confirming that the actuator operated within the specified constraints.

8 RESULTS : SCENARIO B

A constant disturbance torque of 5×10^{-6} N·m – representative of gravity-gradient and aerodynamic disturbance magnitudes reported for small CubeSats in low Earth orbit – was applied throughout the simulation, with all other parameters identical to Scenario A.

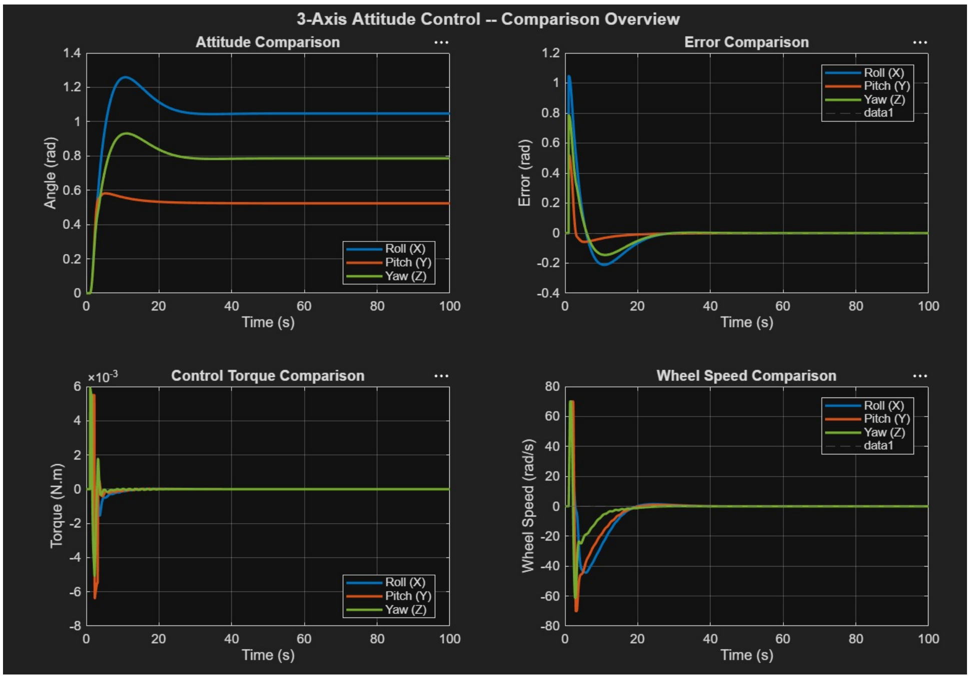

Figure 7: Scenario B: satellite attitude, attitude error, reaction wheel speed, and control torque vs. time.

================= 3-AXIS SIMULATION RESULTS =================

Metric	Roll (X)	Pitch (Y)	Yaw (Z)
Desired Angle (deg)	60.000	30.000	45.000
Rise Time (s)	3.330	1.210	3.460
Peak Time (s)	10.790	5.230	11.120
Overshoot (%)	20.17	11.16	18.54
Settling Time (s)	24.350	18.840	24.700
Steady State Err (rad)	-0.000000	-0.000000	-0.000000
RMS Error (rad)	0.176564	0.071426	0.127097
Max Torque (N.m)	0.005936	0.005936	0.005936
Max Wheel Speed(rad/s)	70.000000	70.000000	70.000000

Figure 8: Scenario B: Performance Metrics (3-AXES)

Discussion: Despite the addition of disturbance torques, the controller maintained stable attitude regulation with negligible steady-state error on all three axes. The rise time, overshoot and settling time remained almost unchanged compared to Scenario A, demonstrating effective disturbance rejection and robustness of the closed-loop controller. The actuator demand also remained within its operating limits, with a maximum torque of 0.0059 N·m and wheel speed limited to 70 rad/s, showing that the disturbances did not cause actuator saturation beyond the designed constraints.

9 CHALLENGES FACED

During the development of the satellite attitude control system, several technical challenges were encountered. One of the primary challenges was extending the model from a simple single-axis system to a fully coupled three-axis rigid-body dynamics model using Euler's rotational equations. Another major challenge was constructing the complete inertia tensor from Fusion 360 CAD data, which required accurate unit conversion and inclusion of the products of inertia.

Considerable effort was also required to design and tune the cascaded PID–PI controller to achieve fast and stable attitude control while minimizing overshoot and settling time. Finally, numerical instability issues, including oscillations, NaN errors, and matrix implementation challenges in Simulink, had to be resolved to obtain accurate and reliable simulation results under different disturbance conditions.

10 CONCLUSION

The developed system successfully three-axis attitude control of a 3U CubeSat using reaction wheels. The final implementation integrates CAD modelling, rigid body dynamics, realistic actuator modelling and closed-loop PID control within a modular Simulink framework. Validation under both disturbance-free and disturbed conditions demonstrates the effectiveness of the proposed control architecture. The addition of a reaction wheel actuation mechanism(PI Controller + DC Motors) further extends the practical applicability of the system and provides scope for future development.